Course Code: EEE-421

Title: Solid State Devices

Session 2021-22: term test only (Tutorial: 01); the final examination is not yet held

Session: 2021-22 **Tutorial: 01 only** **Final: not yet held**

Tutorial 1 (its questions 1(a) to 4 are 7(a) to 7(f))

[Tutorial: 01, Time: 30 minutes, Marks: 20 in all; the 3 marks of 7(a) cover 7(a) and 7(b) together, the 5 marks of 7(d) cover 7(d) and 7(e) together]

7. (a) What is Fermi Dirac distribution? 3

   (b) Explain Ionization Energy.

   (c) Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. 5

   (d) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? 5

   (e) Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature N_c =2.8 $\times$ 10^{19} cm^{-3})

   (f) Silicon atoms, at a concentration of 7 $\times$ 10^{15} cm^{-3}, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x10^6 cm $^{-3}$. 7

      i) Determine the donor and acceptor concentration.
      ii) Is the material n-type or p-type?
      iii) Calculate the electron and hole concentration.
      iv) Determine the position of the Fermi level with respect to E_{Fi}.

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